



Benchmarks Guide

Applies to FinLang v0.6.x

FinLang includes a benchmarking harness to validate performance across dataset sizes, column widths, and ruleset complexities.

This guide shows how to run benchmarks, interpret results, and use them for enterprise capacity planning.

◆ Test Environment

- **CPU:** Intel i7-12700T (12th Gen)
- **RAM:** 48 GB
- **OS:** Windows 11 (64-bit)
- **Python:** 3.11
- **I/O:** PyArrow fast I/O enabled (`--fastio`)

◆ Harnesses

- `bench_finlang_rulesets.py` → tests ruleset-invariance (ruleset size vs performance).
- `bench_finlang_harness.py` → full end-to-end benchmark (rows × cols grid, finale runs).

Run from project root:

```
python -m benchmarks.bench_finlang_harness \
  --mode full-cli \
  --run-fin "finlang --fastio --audit-mode none" \
  --rules examples/rules.demo.fin \
  --include-pack retail,transport,subs \
  --rows 25000,50000,100000,200000 \
  --cols 5,20,35,50 \
  --runs 3 \
  --final-rows 1000000,5000000 \
  --outdir bench_out_final
```

This runs a grid of dataset scales and concludes with finale tests (1M and 5M rows).

◆ Official Results (v0.6.1)

Grid Benchmarks (median of 3 runs)

Rows	Cols	Median Runtime (s)	Median Throughput (rows/s)
25,000	50	1.7	14,806
50,000	50	2.7	18,442

Rows	Cols	Median Runtime (s)	Median Throughput (rows/s)
100,000	50	5.0	20,078
200,000	50	8.7	22,944

Throughput increases as column count decreases (full grid in `bench_results.csv`).

Finale Benchmarks

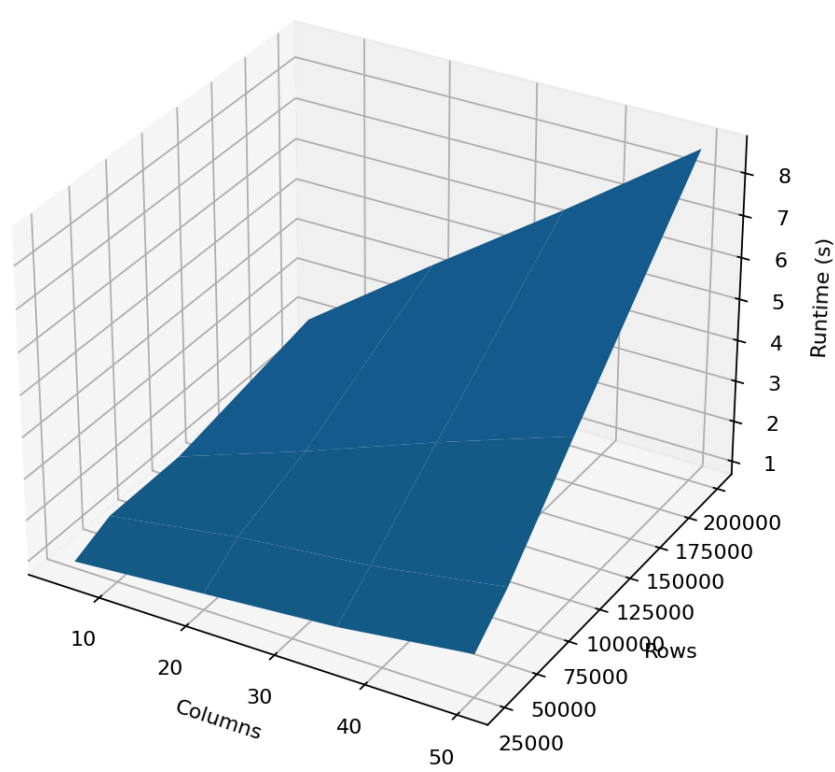
Rows	Cols	Runtime (s)	Throughput (rows/s)
1,000,000	50	41.8	23,900
5,000,000	50	210.3	23,800
5,000,000	20	116.1	43,072
5,000,000	5	37.0	135,214

👉 **Canonical Headline:** 23,800 rows/s @ 5M × 50 cols.

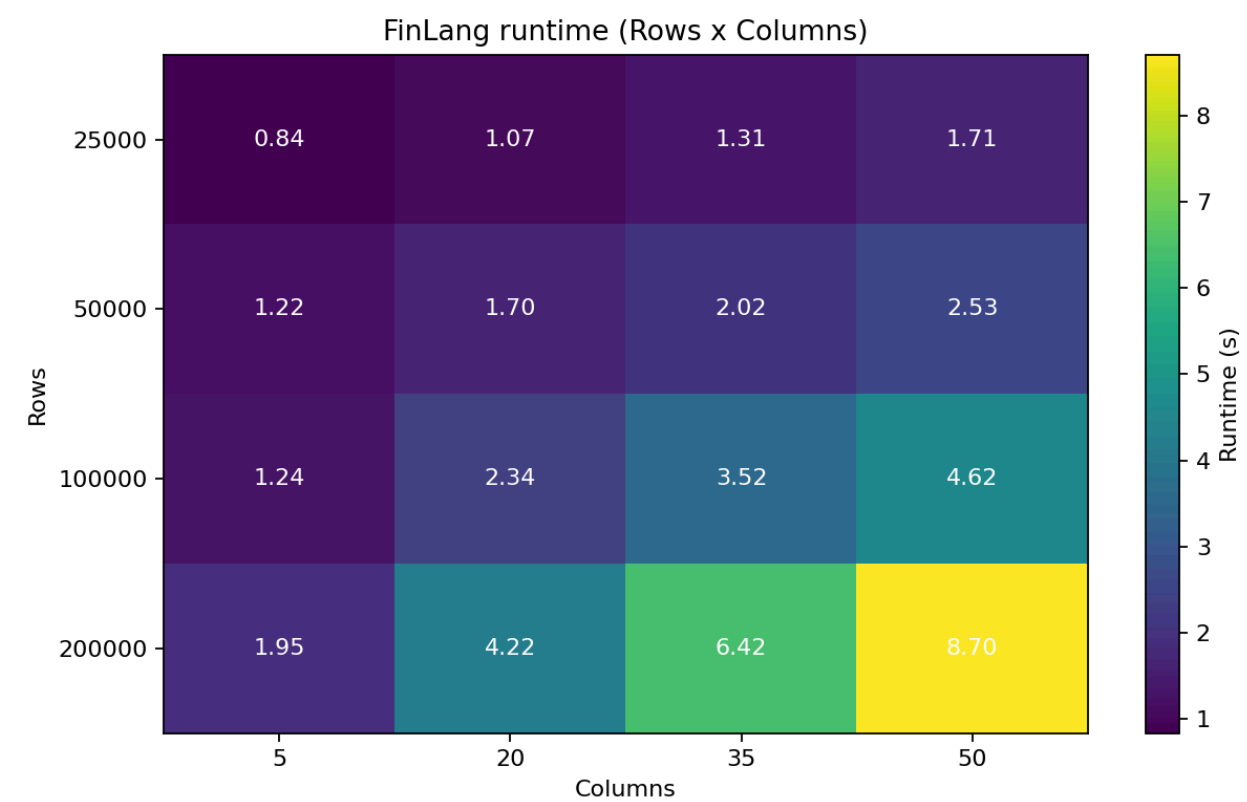
Visualizations

Runtime Surface (Rows × Cols)

FinLang runtime vs Rows x Columns

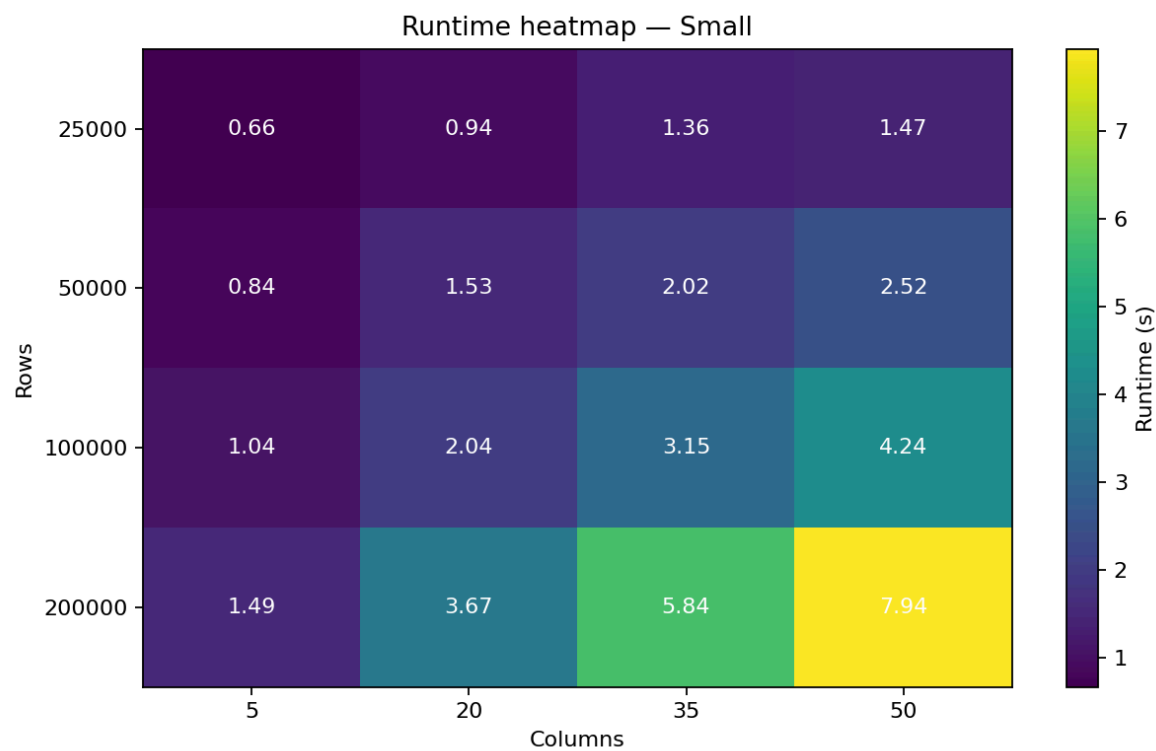


Runtime Heatmap (Rows × Cols)

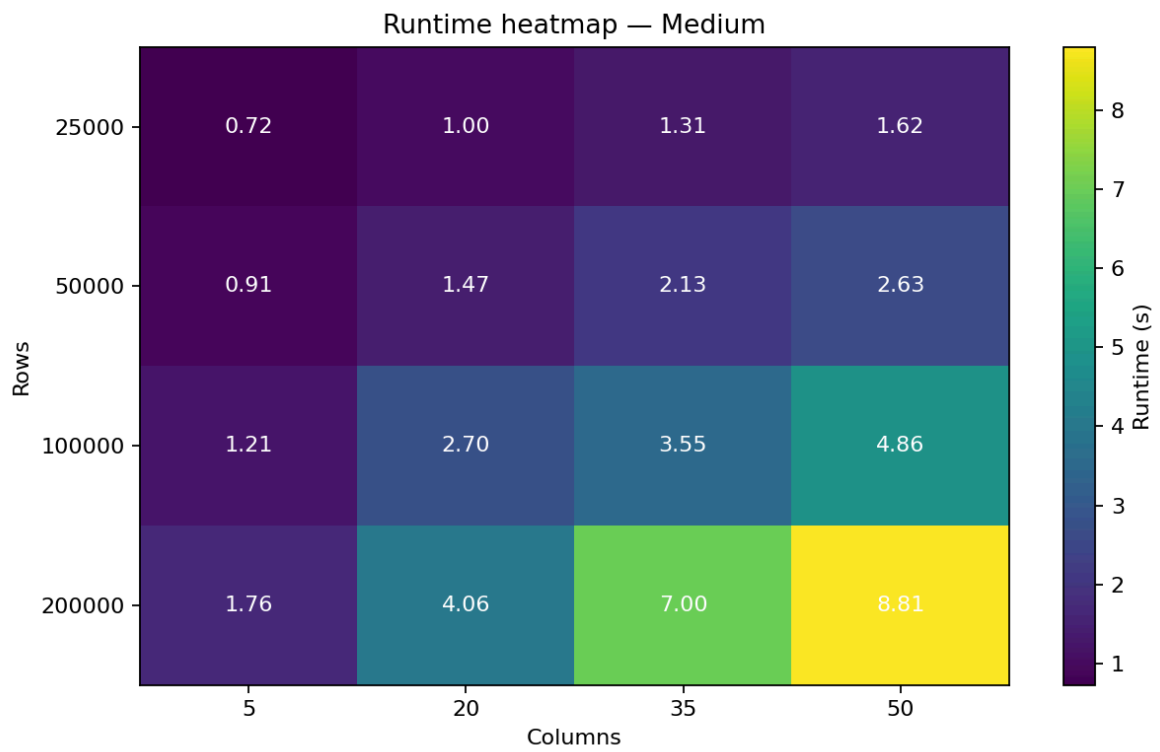


Ruleset Comparison Heatmaps

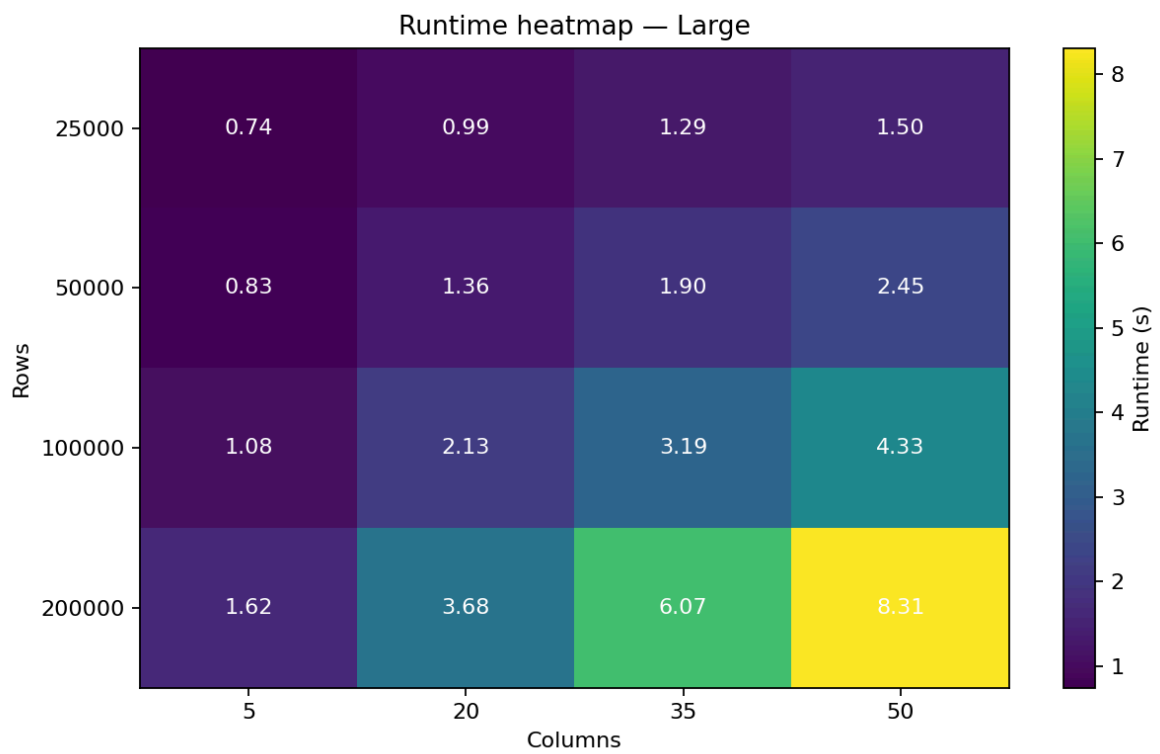
- Small ruleset



- Medium ruleset



- Large ruleset



◆ Findings

Test 1: Performance vs Data Scale

- **Predictable scaling** → runtime grows linearly with rows.
- **Column width adds cost** → more columns = more evaluation overhead.
- **Robust throughput** → >20k rows/s sustained at enterprise scale.

Test 2: Performance vs Rule Complexity

- **Ruleset-invariant** → small, medium, and large rulesets show negligible differences in throughput.
- Confirms FinLang's architectural advantage: rule complexity does not materially impact performance.

◆ Troubleshooting

- **"Command not found"** → use `python -m benchmarks...` or check PATH.
- **Low throughput** → ensure `--fastio` is enabled; otherwise falls back to slower CSV I/O.
- **Audit overhead** → use `--audit-mode none` for benchmarking (fastest mode).
- **Variance across runs** → close background apps, run ≥3 iterations, use median.

◆ Enterprise Integration

- **Regression Testing** → Add benchmark stage in CI/CD; fail builds if throughput drops >10%.
- **Capacity Planning** → Rows/s throughput × SLA gives max dataset size supportable.
- **Compliance Evidence** → Benchmarks prove deterministic scaling; include reports in audit trail.

◆ Cross-References

- [CLI Reference](#) → flags like `--fastio` and `--audit-mode`.
- [Rule Language](#) → rulesets used in tests.
- [Workflows](#) → daily run & growth loop integration.
- [README](#) → headline benchmark highlights.